Can Care Be Automated?

A Data Story About Public Attitudes Toward Robot Caregivers

Motivation and Research Question

This project explores how American adults imagine robot caregivers, especially in the context of elder care. I chose this topic because care robots sit at the intersection of technology, social responsibility, and people relationships. From an HCI and HRI part, a robot caregiver is not only an interface or an intelligent machine. It also becomes part of a care relationship, where questions of trust, companionship, independence, family burden, and human oversight become central.

My initial interest came from the tension between practical usefulness and emotional concern. A robot caregiver may be seen as useful because it can support elder people, reduce pressure on family members, or make care more available. At the same time, people may worry that robotic care could reduce human contact, increase isolation, or turn care into a monitored and less personal system. This makes the topic appropriate for a humanities oriented data visualization project because the issue is not only whether the technology works, but how people imagine the meaning of care when a robot enters the relationship.

The main research question is, Can care be automated without losing the human factor?

More specifically, the dashboard asks how public opinion moves from basic readiness to perceived realism, then to emotional acceptance, willingness, and use of robot caregivers. It also asks which social consequences and written reasons shape acceptance or refusal.

Data Source

The data comes from the Pew Research Center's American Trends Panel Wave 27 survey. This dataset includes survey responses about American adults' attitudes toward robot caregivers. The data package includes the raw SPSS dataset, questionnaire, topline results, and methodology report. The survey wants respondents to imagine a robot caregiver in a home care setting and answer questions about whether the idea is realistic, whether they feel enthusiastic or worried, whether they would use such a robot for themselves or a family member, what consequences they expect, and how they feel about camera based human monitoring. This matched the direction of my original project proposal, where I planned to study public perceptions of care robots through benefits, concerns, willingness, and human oversight.

I cleaned and processed the raw data into several dashboard ready tables. These include respondent level data, a care responsibility heatmap table, a reason summary table, a consequence summary table, and an oversight summary table. I used weighted percentages throughout the dashboard so that the visualizations reflect the survey weighting rather than raw counts only.

Visualization Motivation

The final dashboard is organized as a six chapter data story. I chose this structure because the topic is not well explained by a single bar chart. Public opinion toward robot caregivers is layered. People may find the idea realistic but still not feel enthusiastic. They may recognize benefits while also worrying about isolation. They may accept human monitoring as a safety feature but also see it as a form of surveillance. The dashboard therefore moves from simple interface readiness to more complex social interpretation.

The first chapter, Readiness, uses four summary cards to show familiarity, realism, enthusiasm, and willingness. This establishes the baseline. I used cards rather than a standard bar chart because I wanted readers to experience these measures as stages of public readiness. The cards show that readiness is uneven. People can imagine robot care as realistic before they are emotionally enthusiastic or personally willing to use it.

The second chapter, Pathway, uses a Sankey diagram to show movement from perceived realism, to enthusiasm, and finally willingness to use. This was chosen because the key question is not only how many people support robot caregivers, but how attitudes connect across stages. A Sankey diagram makes it easier to see whether realism becomes enthusiasm and whether enthusiasm becomes willingness. This supports the main argument that acceptance is a pathway, not a single opinion.

The third chapter, Context, uses a heatmap to compare attitudes across different levels of caregiving responsibility. I included this because averages can hide important social differences. Someone who is already doing family caregiving may imagine robot care differently from someone who does not expect to provide care. The heatmap allows readers to compare multiple attitude measures across caregiving groups in one compact matrix.

The fourth chapter, Consequences, uses a promise and anxiety map. I chose this chart because the expected consequences of robot care are not all positive or negative. Some results, such as independence and reduced family burden, are framed as promises. Others, such as isolation or unequal access to human care, are framed as anxieties. This visualization makes the moral tension of robot care visible.

The fifth chapter, Reasons, uses a treemap based on coded written responses. The purpose is to compare why people accept or refuse robot caregivers. The treemap shows that acceptance tends to be organized around practical values such as quality of care, reducing family burden, independence, and availability. Refusal is strongly led by concern over the missing human factor. I chose a treemap because it can show both hierarchy and proportion, making the contrast between acceptance values and refusal values easier to see.

The sixth chapter, Oversight, uses a ternary triangle to show responses to camera monitoring by a human operator. This final chart closes the loop; also, this is my favorite one, because I used this map to analyze the ethical and security issues in my previous ethical HCI research method. If people worry that robot care lacks human presence, human oversight might seem like a solution. However, oversight also introduces questions of surveillance, privacy, and control. The triangle shows whether monitoring makes respondents feel better, worse, or no different. This connects the dashboard back to the larger governance question; robot care is not only about automation, but also about how automated care should be supervised.

Key Findings

The dashboard shows that public opinion toward robot caregivers is not reducible to support or opposition. The first key finding is that perceived realism is higher than willingness to use. This means that people may be able to imagine robot caregivers as technically possible or realistic, while still hesitating to adopt them personally.

The second key finding is that acceptance is filtered through emotional response. The Sankey diagram shows that realism does not automatically translate into enthusiasm, and enthusiasm does not automatically translate into willingness. This matters because it suggests that technical plausibility alone is not enough for public acceptance.

The third key finding is that robot care is interpreted through social context. Caregiving responsibility changes how robot caregivers are imagined. This supports the idea that attitudes toward care robots are not only about the robot itself, but also about the respondent's relationship to family care.

The fourth key finding is that robot caregivers are imagined as both promise and anxiety. They may be associated with independence, reduced family worry, and practical support, but also with isolation and concern that human care may be replaced. This makes robot caregivers a social and ethical issue, not only a product design and technology issue.

The fifth key finding is that acceptance and refusal are driven by different values and thoughts. Acceptance is tied to practical support and independence, while refusal is strongly shaped by the fear that robotic care lacks the human factor. This finding is core to the dashboard's argument, people are not simply judging a machine, but think about whether a machine can stand in for human care.

Outlook

If I had more time and resources, I would expand this project in three directions. First, I would compare this survey with newer datasets to see whether public attitudes have changed as social robots, AI assistants, and home monitoring technologies have become more visible. Second, I would connect the survey analysis with qualitative interviews, especially with older adults, family caregivers, and healthcare workers. This would help explain why certain values, such as independence or human presence, matter so strongly. Third, I would develop the dashboard into an interactive HRI research tool where users could compare different care robot scenarios, such as physical assistance, companionship, medication reminders, and remote monitoring etc.

Overall, this project shows that robot care is not only a technology question. It is a social relationship question. The dashboard moves from public readiness, to attitude pathways, to caregiving context, to social consequences, to reasons for acceptance and refusal, and finally to the governance question of human oversight.

Statement on Collaboration

This was an individual project. I used ChatGPT as a support tool during the development process, mainly to troubleshoot Streamlit and Python implementation issues, revise small pieces of dashboard wording, and check whether the final dashboard clearly addressed the assignment requirements. I also used it to discuss how to present the visual narrative more clearly. The project topic, dataset selection, data cleaning decisions, visualization choices, interpretation of results, design direction, and final submission were completed and reviewed by me.

Links:

- GitHub: https://github.com/williamkij/robot-caregiver-dashboard

- Reference:

Smith, A., & Anderson, M. (2017, October 4). Americans' attitudes toward robot caregivers. Pew Research Center. https://www.pewresearch.org/internet/2017/10/04/americans-attitudes-toward-robot-caregivers/